

Brutal Series - Paper B61: Motion & Time | Heat | Comparing Quantities | Perimeter & Area

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A car covers 150 km in 2 hours 30 minutes at a steady pace. Express its average speed in metres per second (round to 2 decimal places). [\[Motion & Time\]](#)
- (A) 6 m/s
 - (B) 166.67 m/s
 - (C) 41.67 m/s
 - (D) 0.06 m/s
 - (E) 16.67 m/s
 - (F) 1.67 m/s
 - (G) 60 m/s
2. A clinical thermometer shows that a child's mercury rose from a low mark of 35 deg C up to 39.5 deg C. Normal body temperature is 37 deg C. By how much does the reading exceed normal body temperature? [\[Heat\]](#)
- (A) 3.5 deg C
 - (B) 0.5 deg C
 - (C) 2.5 deg C
 - (D) 1.5 deg C
 - (E) 4.5 deg C
 - (F) 39.5 deg C
3. A shopkeeper buys a bag for Rs 450 and sells it for Rs 540. What is the profit percent? [\[Comparing Quantities\]](#)
- (A) 9%
 - (B) 90%
 - (C) 16.67%
 - (D) 20%
 - (E) 25%
 - (F) 20.5%
4. A rectangular garden is 18 m long and 12 m wide. A path 1 m wide runs all around the inside edge. What is the area of the path? [\[Perimeter & Area\]](#)
- (A) 44 m sq
 - (B) 30 m sq
 - (C) 60 m sq
 - (D) 160 m sq
 - (E) 50 m sq
 - (F) 56 m sq

5. A train 180 m long passes a standing pole in exactly 9 seconds. What is the train's speed in km/h? [\[Motion & Time\]](#)
- (A) 90 km/h
 - (B) 18 km/h
 - (C) 36 km/h
 - (D) 20 km/h
 - (E) 5.56 km/h
 - (F) 72 km/h
 - (G) 200 km/h
6. One water sample is heated from 25 deg C to 100 deg C and another from 40 deg C to 100 deg C. Which sample needed the larger temperature rise, and by how much more? [\[Heat\]](#)
- (A) Second, by 15 deg C
 - (B) Both rose equally
 - (C) First, by 75 deg C
 - (D) Second, by 75 deg C
 - (E) First, by 15 deg C
 - (F) First, by 35 deg C
 - (G) First, by 60 deg C
7. A jacket is marked at Rs 2000 and sold at a 15% discount. What is the selling price? [\[Comparing Quantities\]](#)
- (A) Rs 1850
 - (B) Rs 1985
 - (C) Rs 2300
 - (D) Rs 1750
 - (E) Rs 1700
 - (F) Rs 1500
8. A triangular flag is cut so that its base measures 14 cm and the height drawn to that base is 9 cm. Find the area of the flag. [\[Perimeter & Area\]](#)
- (A) 21 cm sq
 - (B) 49 cm sq
 - (C) 23 cm sq
 - (D) 31.5 cm sq
 - (E) 126 cm sq
 - (F) 63 cm sq
 - (G) 98 cm sq
9. A simple pendulum makes 30 complete oscillations in 48 seconds. Find its time period. [\[Motion & Time\]](#)
- (A) 0.625 s
 - (B) 2 s
 - (C) 30 s
 - (D) 16 s
 - (E) 1.5 s
 - (F) 1.6 s
 - (G) 48 s

- 10.** A faulty thermometer reads 2 deg C too high at every point on its scale. It shows 38 deg C for a patient. What is the patient's true temperature? [\[Heat\]](#)
- (A) 2 deg C
 - (B) 40 deg C
 - (C) 36.5 deg C
 - (D) 34 deg C
 - (E) 37 deg C
 - (F) 38 deg C
 - (G) 36 deg C
- 11.** A trader buys an item for Rs 800, marks it up 25%, and then allows a 10% discount on the marked price. What is the final selling price and the profit? [\[Comparing Quantities\]](#)
- (A) Rs 990, no loss
 - (B) Rs 720, a 10% loss
 - (C) Rs 900, a 12.5% profit
 - (D) Rs 1000, a 25% profit
 - (E) Rs 900, a 15% profit
 - (F) Rs 880, a 10% profit
 - (G) Rs 800, no profit
- 12.** A circle has a radius of 21 cm. Using $\pi = \frac{22}{7}$, find its circumference. [\[Perimeter & Area\]](#)
- (A) 66 cm
 - (B) 132 cm
 - (C) 42 cm
 - (D) 138 cm
 - (E) 264 cm
 - (F) 1386 cm
- 13.** A cyclist rides 12 km at 24 km/h, rests for 15 minutes, then rides 8 km at 16 km/h. How long is the whole trip? [\[Motion & Time\]](#)
- (A) 1 h
 - (B) 45 min
 - (C) 1 h 15 min
 - (D) 50 min
 - (E) 2 h
 - (F) 1 h 30 min
 - (G) 1 h 45 min
- 14.** Left in the sun, a dark car's cabin rose from 30 deg C to 54 deg C while a white car's cabin rose from 30 deg C to 42 deg C. The dark car's temperature rise was how many times the white car's rise? [\[Heat\]](#)
- (A) 1.5 times
 - (B) 0.5 times
 - (C) 3 times
 - (D) 2 times
 - (E) 1.3 times
 - (F) 54 times

15. Find the simple interest on Rs 5000 borrowed at 8% per annum for 3 years. [\[Comparing Quantities\]](#)

- (A) Rs 1200
- (B) Rs 1500
- (C) Rs 1300
- (D) Rs 400
- (E) Rs 6200
- (F) Rs 120
- (G) Rs 4000

16. A circle has a radius of 7 cm. Using $\pi = 22/7$, find its area. [\[Perimeter & Area\]](#)

- (A) 77 cm sq
- (B) 154 cm sq
- (C) 22 cm sq
- (D) 100 cm sq
- (E) 44 cm sq
- (F) 49 cm sq

17. A bus travels the first 60 km of a route at 30 km/h and the next 60 km at 60 km/h. What is its average speed for the whole journey? [\[Motion & Time\]](#)

- (A) 40 km/h
- (B) 90 km/h
- (C) 50 km/h
- (D) 120 km/h
- (E) 45 km/h
- (F) 30 km/h

18. A room heater warms an entire room even though the air is mostly still to begin with. Which statement best explains how the warmth spreads through the room air? [\[Heat\]](#)

- (A) Cold air rises and hot air sinks through the room
- (B) Warm air rises and cooler air sinks, setting up convection currents
- (C) Sea breezes carry the heat across the room
- (D) Heat passes by conduction because air is a good conductor
- (E) The walls conduct heat faster than the air
- (F) Only radiation warms the room, with no air movement
- (G) Heat spreads because air boils and evaporates

19. Rs 2400 is deposited at 5% per annum simple interest for 2 years. What total amount is in the account at the end? [\[Comparing Quantities\]](#)

- (A) Rs 2160
- (B) Rs 2880
- (C) Rs 2400
- (D) Rs 2645
- (E) Rs 240
- (F) Rs 2640

20. A parallelogram has a base of 12 cm and a corresponding height of 7.5 cm. What is its area? [\[Perimeter & Area\]](#)
- (A) 39 cm sq
 - (B) 19.5 cm sq
 - (C) 60 cm sq
 - (D) 19.5 cm
 - (E) 45 cm sq
 - (F) 82.5 cm sq
 - (G) 90 cm sq
21. A car moves at a steady 90 km/h. How far, in metres, does it travel in 4 seconds? [\[Motion & Time\]](#)
- (A) 100 m
 - (B) 25 m
 - (C) 1000 m
 - (D) 400 m
 - (E) 6.25 m
 - (F) 90 m
 - (G) 360 m
22. A laboratory thermometer's scale runs from 0 deg C to 100 deg C over a length of 20 cm. How many degrees Celsius does each centimetre of the scale represent? [\[Heat\]](#)
- (A) 2 deg C per cm
 - (B) 0.2 deg C per cm
 - (C) 5 deg C per cm
 - (D) 100 deg C per cm
 - (E) 50 deg C per cm
 - (F) 10 deg C per cm
23. In a class of 40 students the ratio of boys to girls is 3:2. How many girls are there? [\[Comparing Quantities\]](#)
- (A) 8
 - (B) 16
 - (C) 2
 - (D) 20
 - (E) 18
 - (F) 32
 - (G) 24
24. A square has a perimeter of 56 cm. What is its area? [\[Perimeter & Area\]](#)
- (A) 784 cm sq
 - (B) 49 cm sq
 - (C) 14 cm sq
 - (D) 112 cm sq
 - (E) 56 cm sq
 - (F) 196 cm sq
 - (G) 224 cm sq

- 25.** A car's odometer reads 56,128 km at 8:00 a.m. and 56,608 km at 2:00 p.m. on a long drive. What was its average speed? [\[Motion & Time\]](#)
- (A) 40 km/h
 - (B) 80 km/h
 - (C) 240 km/h
 - (D) 60 km/h
 - (E) 96 km/h
 - (F) 160 km/h
 - (G) 480 km/h
- 26.** Water in a pot heats up at a steady rate of 5 deg C every minute. How long will it take to warm the water from 30 deg C to 100 deg C? [\[Heat\]](#)
- (A) 35 min
 - (B) 13 min
 - (C) 20 min
 - (D) 14 s
 - (E) 70 min
 - (F) 14 min
 - (G) 12 min
- 27.** In an exam, 60% of the students passed. Of those who passed, 25% earned a distinction. What percent of all the students earned a distinction? [\[Comparing Quantities\]](#)
- (A) 45%
 - (B) 12%
 - (C) 25%
 - (D) 85%
 - (E) 60%
 - (F) 35%
 - (G) 15%
- 28.** A square field has an area of 169 m sq. What is its perimeter? [\[Perimeter & Area\]](#)
- (A) 42.25 m
 - (B) 13 m
 - (C) 676 m
 - (D) 48 m
 - (E) 52 m
 - (F) 169 m
 - (G) 26 m
- 29.** Sound travels at 340 m/s in air. A child sees lightning and hears the thunder 5 seconds later. How far away is the storm, in kilometres? [\[Motion & Time\]](#)
- (A) 1700 km
 - (B) 3.4 km
 - (C) 5 km
 - (D) 0.34 km
 - (E) 1.7 km
 - (F) 17 km
 - (G) 68 km

30. Engineers leave small gaps between consecutive steel rails on a railway line. Why are these gaps deliberately left? [\[Heat\]](#)

- (A) To save steel and reduce cost
- (B) To let rainwater drain off the track
- (C) To give train wheels a better grip
- (D) Because metals shrink when they get hot
- (E) To let electric current pass between rails
- (F) To allow the rails to expand in summer heat without bending
- (G) To allow the rails to contract when heated

31. A number, when increased by 20%, becomes 360. What was the original number? [\[Comparing Quantities\]](#)

- (A) 300
- (B) 340
- (C) 320
- (D) 288
- (E) 250
- (F) 432
- (G) 360

32. A rectangle has an area of 240 cm sq and a length of 20 cm. What is its perimeter? [\[Perimeter & Area\]](#)

- (A) 12 cm
- (B) 32 cm
- (C) 240 cm
- (D) 64 cm
- (E) 60 cm
- (F) 44 cm
- (G) 52 cm

33. A car's speed rises steadily from 36 km/h to 72 km/h. By how many metres per second did its speed increase? [\[Motion & Time\]](#)

- (A) 20 m/s
- (B) 36 m/s
- (C) 18 m/s
- (D) 1 m/s
- (E) 10 m/s
- (F) 5 m/s

34. Two litres of water at 80 deg C are mixed with two litres of water at 20 deg C in an insulated jar. What is the final temperature of the mixture? [\[Heat\]](#)

- (A) 80 deg C
- (B) 100 deg C
- (C) 40 deg C
- (D) 30 deg C
- (E) 50 deg C
- (F) 45 deg C

- 35.** An employee's monthly salary of Rs 25,000 is raised by 12%. What is the new salary? [\[Comparing Quantities\]](#)
- (A) Rs 28000
 - (B) Rs 22000
 - (C) Rs 28120
 - (D) Rs 28200
 - (E) Rs 37000
 - (F) Rs 26000
 - (G) Rs 3000
- 36.** A large rectangle measures 20 cm by 15 cm. A small 5 cm by 5 cm square is cut out from one corner. What area remains? [\[Perimeter & Area\]](#)
- (A) 275 cm
 - (B) 25 cm sq
 - (C) 275 cm sq
 - (D) 250 cm sq
 - (E) 325 cm sq
 - (F) 290 cm sq
 - (G) 300 cm sq
- 37.** Two stations are 240 km apart. A train leaves the first at 9:00 a.m. and runs steadily at 60 km/h. At what clock time does it reach the second station? [\[Motion & Time\]](#)
- (A) 11:00 a.m.
 - (B) 9:04 a.m.
 - (C) 3:00 p.m.
 - (D) 1:00 p.m.
 - (E) 12:00 noon
 - (F) 4:00 p.m.
 - (G) 2:00 p.m.
- 38.** Three litres of water at 60 deg C are mixed with one litre of water at 20 deg C. Assuming no heat is lost, what is the final temperature? [\[Heat\]](#)
- (A) 40 deg C
 - (B) 35 deg C
 - (C) 55 deg C
 - (D) 50 deg C
 - (E) 45 deg C
 - (F) 50.5 deg C
- 39.** Express the fraction $\frac{3}{8}$ as a percentage. [\[Comparing Quantities\]](#)
- (A) 62.5%
 - (B) 3.8%
 - (C) 0.375%
 - (D) 8.3%
 - (E) 30%
 - (F) 37.5%

40. A triangle has an area of 84 cm sq and a base of 24 cm. What is its height? [\[Perimeter & Area\]](#)

- (A) 6 cm
- (B) 14 cm
- (C) 7 cm
- (D) 84 cm
- (E) 3.5 cm sq
- (F) 3.5 cm

41. A pendulum has a time period of 2 seconds. How many complete oscillations does it make in one minute? [\[Motion & Time\]](#)

- (A) 15
- (B) 30
- (C) 0.5
- (D) 120
- (E) 60
- (F) 45
- (G) 2

42. Places near the sea have a milder, more even climate than places far inland. Which fact best explains this moderating effect of the sea? [\[Heat\]](#)

- (A) Water heats up and cools down very quickly
- (B) Sea breezes only ever blow at night
- (C) Land stores far more heat than the sea
- (D) Water boils easily and releases heat
- (E) Land is a much better conductor than water
- (F) Water heats up and cools down slowly, so it moderates the temperature

43. A price is first reduced by 20% and then the new price is reduced by a further 10%. What single discount is equivalent to these two together? [\[Comparing Quantities\]](#)

- (A) 25%
- (B) 28%
- (C) 26%
- (D) 72%
- (E) 18%
- (F) 32%

44. A wheel's circumference is 88 cm. Using $\pi = \frac{22}{7}$, what is its radius? [\[Perimeter & Area\]](#)

- (A) 56 cm
- (B) 22 cm
- (C) 14 cm
- (D) 88 cm
- (E) 7 cm
- (F) 28 cm

- 45.** An athlete runs 4 laps of a 400 m track in 6 minutes 40 seconds. What is her average speed in metres per second? [\[Motion & Time\]](#)
- (A) 2.4 m/s
 - (B) 16 m/s
 - (C) 0.25 m/s
 - (D) 240 m/s
 - (E) 6.4 m/s
 - (F) 4 m/s
 - (G) 400 m/s
- 46.** Heat from the Sun reaches the Earth across the empty space of the solar system. By which mode of heat transfer does most of this solar heat travel? [\[Heat\]](#)
- (A) Radiation
 - (B) Conduction and convection together
 - (C) By evaporation
 - (D) Conduction
 - (E) Convection
 - (F) By sea breezes
 - (G) By reflection
- 47.** A sum of Rs 4000 grows to Rs 4960 in 2 years under simple interest. What is the rate of interest per annum? [\[Comparing Quantities\]](#)
- (A) 960%
 - (B) 12%
 - (C) 10%
 - (D) 12.4%
 - (E) 24%
 - (F) 8%
 - (G) 6%
- 48.** A rectangular garden 25 m by 15 m has a path 2 m wide running all around it on the outside. What is the area of the path? [\[Perimeter & Area\]](#)
- (A) 176 m sq
 - (B) 551 m sq
 - (C) 160 m sq
 - (D) 88 m sq
 - (E) 200 m sq
 - (F) 375 m sq
- 49.** A toy car travels 20 m in 8 s and then 30 m in 12 s without stopping. What is its average speed for the whole run? [\[Motion & Time\]](#)
- (A) 2.5 m/s
 - (B) 3 m/s
 - (C) 2.45 m/s
 - (D) 50 m/s
 - (E) 1.25 m/s
 - (F) 20 m/s

- 50.** A weather station records a daytime high of 41 deg C and a night-time low of 26 deg C. What was the midpoint (mean) temperature of that day? [\[Heat\]](#)
- (A) 30 deg C
 - (B) 34 deg C
 - (C) 15 deg C
 - (D) 33 deg C
 - (E) 67 deg C
 - (F) 13.5 deg C
 - (G) 33.5 deg C
- 51.** An article is sold for Rs 575 at a profit of 15%. What was its cost price? [\[Comparing Quantities\]](#)
- (A) Rs 560
 - (B) Rs 525
 - (C) Rs 488.75
 - (D) Rs 575
 - (E) Rs 490
 - (F) Rs 660.25
 - (G) Rs 500
- 52.** A right-angled triangle has legs of 9 cm and 12 cm and a hypotenuse of 15 cm. What is its area? [\[Perimeter & Area\]](#)
- (A) 54 cm sq
 - (B) 108 cm sq
 - (C) 36 cm sq
 - (D) 18 cm sq
 - (E) 90 cm sq
 - (F) 54 cm
 - (G) 67.5 cm sq
- 53.** A moving object has a steady speed of 15 m/s. Express this speed in kilometres per hour. [\[Motion & Time\]](#)
- (A) 60 km/h
 - (B) 15 km/h
 - (C) 54 km/h
 - (D) 150 km/h
 - (E) 5.4 km/h
 - (F) 27 km/h
- 54.** Food cooks faster in a pressure cooker than in an open pot. What is the main scientific reason for this? [\[Heat\]](#)
- (A) Steam inside is colder than boiling water
 - (B) Water inside boils at a lower temperature
 - (C) The pressure lowers the boiling point of water
 - (D) The cooker adds extra heat energy from outside
 - (E) Pressure cools the food so it sets faster
 - (F) The cooker removes all the air around the food
 - (G) Higher pressure raises the boiling point of water above 100 deg C

55. Which is the better buy: 2 kg of rice for Rs 90, or 3 kg of the same rice for Rs 120? [\[Comparing Quantities\]](#)

- (A) 3 kg for Rs 120, at Rs 30 per kg
- (B) 2 kg for Rs 90, at Rs 40 per kg
- (C) 3 kg for Rs 120, at Rs 40 per kg
- (D) Both cost the same per kg
- (E) 2 kg for Rs 90, at Rs 45 per kg
- (F) It cannot be decided

56. A floor measures 4 m by 3 m and is to be covered with square tiles of side 20 cm. How many tiles are needed? [\[Perimeter & Area\]](#)

- (A) 150
- (B) 300
- (C) 60
- (D) 250
- (E) 1200
- (F) 30
- (G) 3000

57. The second hand of a clock sweeps a full circle in 60 seconds. Through how many degrees does it turn in 15 seconds? [\[Motion & Time\]](#)

- (A) 15 deg
- (B) 30 deg
- (C) 90 deg
- (D) 360 deg
- (E) 180 deg
- (F) 6 deg
- (G) 45 deg

58. In sunshine, a black water tank's temperature climbs 2 deg C each minute while a shiny tank climbs 0.5 deg C each minute. Both start at 25 deg C. After 10 minutes, by how much do their temperatures differ?

[\[Heat\]](#)

- (A) 20 deg C
- (B) 15 deg C
- (C) 12.5 deg C
- (D) 10 deg C
- (E) 30 deg C
- (F) 1.5 deg C

59. A town has a population of 8000 that grows by 5% in one year. What is the population after that year?

[\[Comparing Quantities\]](#)

- (A) 8500
- (B) 8400
- (C) 12000
- (D) 7600
- (E) 8450
- (F) 8040

- 60.** Two circles share the same centre, with radii of 14 cm and 7 cm. Using $\pi = \frac{22}{7}$, find the area of the ring between them. [\[Perimeter & Area\]](#)
- (A) 308 cm sq
 - (B) 770 cm sq
 - (C) 147 cm sq
 - (D) 462 cm sq
 - (E) 616 cm sq
 - (F) 462 cm
 - (G) 154 cm sq
- 61.** A car travels at 50 km/h for 2 hours and then at 70 km/h for 3 hours on a highway. What total distance does it cover? [\[Motion & Time\]](#)
- (A) 120 km
 - (B) 250 km
 - (C) 610 km
 - (D) 210 km
 - (E) 310 km
 - (F) 300 km
- 62.** We feel warmer in woollen clothes in winter. What is the correct reason wool keeps us warm? [\[Heat\]](#)
- (A) Trapped air is a good conductor that warms us
 - (B) Wool produces heat by friction
 - (C) Wool traps air, a poor conductor, which slows heat loss from the body
 - (D) Wool reflects body heat back as radiation
 - (E) Wool absorbs sunlight to warm us
 - (F) Wool is a good conductor of heat
- 63.** A man spends 60% of his monthly income and saves the rest, which comes to Rs 4800. What is his monthly income? [\[Comparing Quantities\]](#)
- (A) Rs 2880
 - (B) Rs 12000
 - (C) Rs 6000
 - (D) Rs 7680
 - (E) Rs 12400
 - (F) Rs 8000
 - (G) Rs 19200
- 64.** A rectangular field measures 40 m by 30 m. It is fenced all around at a cost of Rs 25 per metre. What is the total cost of fencing? [\[Perimeter & Area\]](#)
- (A) Rs 30000
 - (B) Rs 7000
 - (C) Rs 2800
 - (D) Rs 140
 - (E) Rs 3600
 - (F) Rs 1750
 - (G) Rs 3500

- 65.** A bus must average a certain speed to be on time. After covering 150 km in the first 3 hours, 150 km still remain and only 2 hours are left. What speed must it now keep to arrive on time? [\[Motion & Time\]](#)
- (A) 60 km/h
 - (B) 45 km/h
 - (C) 50 km/h
 - (D) 30 km/h
 - (E) 75 km/h
 - (F) 150 km/h
 - (G) 100 km/h
- 66.** A patient's temperature is 40 deg C, while normal body temperature is 37 deg C. By what percentage is 40 deg C above the normal value (round to the nearest whole percent)? [\[Heat\]](#)
- (A) about 13%
 - (B) about 8%
 - (C) about 7.5%
 - (D) about 92%
 - (E) about 10%
 - (F) about 3%
- 67.** A phone costs Rs 15,000 before tax, and 18% GST is added at the counter. What is the total amount to be paid? [\[Comparing Quantities\]](#)
- (A) Rs 17800
 - (B) Rs 2700
 - (C) Rs 17000
 - (D) Rs 18000
 - (E) Rs 17700
 - (F) Rs 12300
- 68.** A square has a side of 16 cm and a rectangle measures 20 cm by 12 cm. Which has the larger area, and by how much? [\[Perimeter & Area\]](#)
- (A) The square, by 16 cm sq
 - (B) The rectangle, by 16 cm sq
 - (C) They are equal
 - (D) The rectangle, by 240 cm sq
 - (E) The square, by 256 cm sq
 - (F) The square, by 4 cm sq
- 69.** A pendulum makes 25 complete oscillations in one minute. Find its time period. [\[Motion & Time\]](#)
- (A) 0.417 s
 - (B) 2.5 s
 - (C) 25 s
 - (D) 4.2 s
 - (E) 60 s
 - (F) 2.4 s
 - (G) 1.5 s

70. Heat is applied at one end of two metal rods, and wax fixed at the far end melts in 20 s on rod A and in 50 s on rod B. Which rod is the better conductor, and how many times faster does it carry heat (by melt time)? [\[Heat\]](#)

- (A) Rod A, 0.4 times as fast
- (B) Rod A, 70 times faster
- (C) Rod B, 2 times faster
- (D) Rod B, 2.5 times faster
- (E) Both conduct equally
- (F) Rod A, 2.5 times faster
- (G) Rod A, 30 times faster

71. A recipe mixes sugar and flour in the ratio 2:5. To use 750 g of flour, how much sugar is needed?

[\[Comparing Quantities\]](#)

- (A) 525 g
- (B) 150 g
- (C) 1875 g
- (D) 600 g
- (E) 375 g
- (F) 300 g
- (G) 250 g

72. A parallelogram has an area of 96 cm sq and a base of 16 cm. What is the corresponding height?

[\[Perimeter & Area\]](#)

- (A) 48 cm
- (B) 6 cm
- (C) 12 cm
- (D) 8 cm
- (E) 96 cm
- (F) 6 cm sq
- (G) 3 cm

73. An object covers equal distances in equal times, moving 12 m in every 4 seconds. How far will it travel in 10 seconds? [\[Motion & Time\]](#)

- (A) 3 m
- (B) 12 m
- (C) 48 m
- (D) 120 m
- (E) 30 m
- (F) 0.3 m
- (G) 40 m

74. At a coast the sea breeze blows from 10 a.m. to 7 p.m. For how many hours does it blow, and what fraction of a full day is that? [\[Heat\]](#)

- (A) 9 hours, which is $\frac{1}{4}$ of a day
- (B) 9 hours, which is $\frac{3}{8}$ of a day
- (C) 7 hours, which is $\frac{7}{24}$ of a day
- (D) 9 hours, which is $\frac{9}{12}$ of a day
- (E) 17 hours, which is $\frac{17}{24}$ of a day
- (F) 9 hours, which is $\frac{1}{3}$ of a day
- (G) 10 hours, which is $\frac{5}{12}$ of a day

75. A student scored 348 out of 600 in an exam. How many more marks were needed to reach exactly 70%? [\[Comparing Quantities\]](#)

- (A) already above 70%
- (B) 120 more marks
- (C) 252 more marks
- (D) 70 more marks
- (E) 72 more marks
- (F) 12 more marks
- (G) 42 more marks

76. A piece of wire is bent into a square of side 11 cm, then straightened and bent into a circle. Using $\pi = \frac{22}{7}$, what is the area of the circle? [\[Perimeter & Area\]](#)

- (A) 121 cm sq
- (B) 44 cm sq
- (C) 154 cm
- (D) 616 cm sq
- (E) 154 cm sq
- (F) 49 cm sq

77. A bicycle wheel has a radius of 35 cm. Using $\pi = \frac{22}{7}$, how many full turns does it make while the cycle covers 1100 m? [\[Motion & Time\]](#)

- (A) 50
- (B) 5000
- (C) 250
- (D) 1000
- (E) 242
- (F) 5
- (G) 500

78. A thermometer reads 0 deg C in melting ice and 100 deg C in steam, where its mercury column is 25 cm long. While measuring a warm liquid, the column stands 6 cm above the ice mark. What is the liquid's temperature? [\[Heat\]](#)

- (A) 4 deg C
- (B) 100 deg C
- (C) 60 deg C
- (D) 6 deg C
- (E) 20 deg C
- (F) 150 deg C
- (G) 24 deg C

79. After a 20% discount, a television sells for Rs 16,000. What was its marked price? [\[Comparing Quantities\]](#)

- (A) Rs 19200
- (B) Rs 20000
- (C) Rs 16800
- (D) Rs 20800
- (E) Rs 3200
- (F) Rs 18000

80. From a rectangular card 40 cm by 28 cm, the largest possible circle is cut out. Using $\pi = \frac{22}{7}$, what is the area of that circle? [\[Perimeter & Area\]](#)

- (A) 196 cm sq
- (B) 88 cm sq
- (C) 1232 cm sq
- (D) 616 cm
- (E) 616 cm sq
- (F) 154 cm sq

81. Two cars start 200 km apart and drive straight toward each other, one at 40 km/h and the other at 60 km/h. After how long do they meet? [\[Motion & Time\]](#)

- (A) 4 h
- (B) 3.33 h
- (C) 1 h
- (D) 2 h
- (E) 2.5 h
- (F) 5 h
- (G) 20 h

82. A desert reaches 48 deg C by day and falls to 10 deg C at night because sand heats and cools quickly. By how many degrees does the temperature drop overnight? [\[Heat\]](#)

- (A) 58 deg C
- (B) 38 deg C
- (C) 19 deg C
- (D) 48 deg C
- (E) 10 deg C
- (F) 40 deg C
- (G) 29 deg C

83. In how many years will Rs 6000 earn Rs 1440 in simple interest at a rate of 6% per annum? [\[Comparing Quantities\]](#)

- (A) 6 years
- (B) 3 years
- (C) 4.4 years
- (D) 2 years
- (E) 5 years
- (F) 4 years
- (G) 24 years

- 84.** A plot of land has an area of 2 hectares. Given that 1 hectare = 10,000 m sq, express its area in square metres. [\[Perimeter & Area\]](#)
- (A) 100000 m sq
 - (B) 20000 m sq
 - (C) 200000 m sq
 - (D) 2 m sq
 - (E) 20 m sq
 - (F) 2000 m sq
- 85.** A man walks at a steady 5 km/h and reaches his office in 36 minutes. How far is his office? [\[Motion & Time\]](#)
- (A) 0.6 km
 - (B) 30 km
 - (C) 1.8 km
 - (D) 180 km
 - (E) 3 km
 - (F) 2.5 km
 - (G) 3.6 km
- 86.** After two minutes in a cup of hot tea, a metal spoon's handle feels hot but a plastic spoon's handle stays cool. What best explains this difference? [\[Heat\]](#)
- (A) Metal conducts heat from the tea to the handle; plastic is a poor conductor
 - (B) Plastic conducts heat faster than metal
 - (C) Both spoons conduct heat equally well
 - (D) The handle radiates the tea's light as heat
 - (E) Metal reflects the heat away from itself
 - (F) The tea is hotter near the plastic spoon
 - (G) Plastic makes heat by rubbing the tea
- 87.** It is given that $A : B = 2 : 3$ and $B : C = 4 : 5$. What is the ratio $A : C$? [\[Comparing Quantities\]](#)
- (A) 12 : 15
 - (B) 8 : 12
 - (C) 5 : 8
 - (D) 2 : 3
 - (E) 10 : 12
 - (F) 2 : 5
 - (G) 8 : 15
- 88.** Inside a rectangle 16 cm long and 10 cm wide, a triangle is drawn whose base is the full 16 cm length and whose apex touches the opposite side. What is the triangle's area? [\[Perimeter & Area\]](#)
- (A) 20 cm sq
 - (B) 80 cm
 - (C) 80 cm sq
 - (D) 53 cm sq
 - (E) 40 cm sq
 - (F) 160 cm sq
 - (G) 26 cm sq

89. Object P moves 100 m in 25 s. Object Q moves at 72 km/h. Which is faster, and by how much in metres per second? [\[Motion & Time\]](#)

- (A) Q is faster by 68 m/s
- (B) Both move equally fast
- (C) P is faster by 16 m/s
- (D) Q is faster by 16 m/s
- (E) P is faster by 4 m/s
- (F) Q is faster by 1.6 m/s
- (G) Q is faster by 21 m/s

90. A heater warms a room at a steady 3 deg C every 10 minutes. If the room starts at 16 deg C, how long will it take to reach 28 deg C? [\[Heat\]](#)

- (A) 20 min
- (B) 50 min
- (C) 40 min
- (D) 12 min
- (E) 30 min
- (F) 36 min
- (G) 28 min

91. A bike bought for Rs 40,000 is sold for Rs 34,000. What is the loss percent? [\[Comparing Quantities\]](#)

- (A) 17.6%
- (B) 85%
- (C) 15.5%
- (D) 12%
- (E) 15%
- (F) 20%
- (G) 6%

92. A room measures 6 m by 5 m and is to be carpeted at Rs 120 per square metre. What is the total cost of the carpet? [\[Perimeter & Area\]](#)

- (A) Rs 360
- (B) Rs 30
- (C) Rs 1320
- (D) Rs 2640
- (E) Rs 3000
- (F) Rs 4320
- (G) Rs 3600

93. A sprinter runs 200 m in 25 s. To set a record she must average 10 m/s over the same distance. By how many seconds must she improve her time? [\[Motion & Time\]](#)

- (A) 15 s
- (B) 2 s
- (C) 5 s
- (D) 20 s
- (E) 10 s
- (F) 8 s
- (G) 2.5 s

94. A clinical thermometer has a narrow kink (a constriction) in its tube just above the bulb. What is the purpose of this kink? [\[Heat\]](#)

- (A) It makes the thermometer read faster
- (B) It stops the mercury from slipping back, so the reading holds until shaken down
- (C) It lets the thermometer read in Fahrenheit
- (D) It increases the top of the range to 110 deg C
- (E) It cools the mercury after a reading
- (F) It is only there for decoration

95. A worker's salary is first raised by 10% and then, in a downturn, cut by 10% of the new figure. What is the net change in the salary? [\[Comparing Quantities\]](#)

- (A) No change at all
- (B) 2% decrease
- (C) 0.1% decrease
- (D) 1% decrease
- (E) 1% increase
- (F) 20% decrease

96. If each side of a square is increased by 10%, by what percent does its area increase? [\[Perimeter & Area\]](#)

- (A) 11%
- (B) 10%
- (C) 21%
- (D) 121%
- (E) 20%
- (F) 1.21%

97. A student walks 3 km to school at 6 km/h and walks the same 3 km home at 4 km/h. What is the average speed for the whole there-and-back trip? [\[Motion & Time\]](#)

- (A) 6 km/h
- (B) 2.4 km/h
- (C) 10 km/h
- (D) 4.8 km/h
- (E) 4.5 km/h
- (F) 5.5 km/h
- (G) 5 km/h

98. Hot water at 70 deg C and cold water at 10 deg C are mixed in the ratio 1 part hot to 2 parts cold. Assuming no heat loss, what is the final temperature? [\[Heat\]](#)

- (A) 60 deg C
- (B) 50 deg C
- (C) 30 deg C
- (D) 40 deg C
- (E) 35 deg C
- (F) 26.7 deg C

99. A shopkeeper sells two clocks at Rs 990 each. On one she makes a 10% profit and on the other a 10% loss. Overall, does she gain or lose, and by how much? [\[Comparing Quantities\]](#)

- (A) A profit of Rs 20
- (B) A profit of 1%
- (C) A loss of Rs 40
- (D) A loss of Rs 100
- (E) Neither profit nor loss
- (F) A loss of 2%
- (G) A loss of Rs 20 (1%)

100. A rectangular plot 50 m by 40 m has two crossing paths each 5 m wide: one runs the full 50 m length and the other the full 40 m width, meeting in the middle. What is the total area of the paths? [\[Perimeter & Area\]](#)

- (A) 225 m sq
- (B) 425 m
- (C) 425 m sq
- (D) 250 m sq
- (E) 450 m sq
- (F) 475 m sq